Tables

Table 1. Pulse sequences provided.

Purpose	Scanner	Folder*	Filenames
CEST-MRF of L-arginine (3 ppm)[30,73]	Preclinical (Bruker) / Clinical scanners	cest_mrf	cest_mrf_preclinical.txt cest_mrf_preclinical.seq cest_mrf_clinical.seq
CEST-weighted imaging[102]	Preclinical (Bruker)	cest_weighted	cest_weighted.txt
CEST-MRF Brain imaging (2D EPI)[41]	Preclinical (Bruker) / Clinical	2depi_cest_mrf	2D_EPI_CEST.seq 2D_EPI_CEST.txt
Deep learning optimized CEST-MRF human brain imaging protocol[67]	Preclinical (Bruker) / Clinical scanners	scone	scone.txt scone.seq
Deep learning optimized CEST MRF acquisition of BSA (rNOE, amine, and amide), semisolid MT mouse brain, and pCR (all at 9.4T)[47]	Preclinical (Bruker)	autocest	212_rNOE_bsa_9.4T.txt 216_bsa_amine_2p75ppm_9.4T.txt 222_bsa_amide_3p5ppm_9.4T.txt 292_mt_in_vivo_mouse_9.4T.txt 311_pCr_2p6ppm_9.4T.txt
Semisolid MT MRF protocol used in the unsupervised pipeline[40]	Preclinical (Bruker) / Clinical scanners	mtc_mrf_unsupervised	mt_unsuper.txt mt_unsuper.seq
Deep learning optimized Semisolid MT MRF protocols[77]	Preclinical (Bruker) / Clinical scanners	loas_mtc_mrf	loas_mtc_10.txt loas_mtc_10.seq loas_mtc_40.txt loas_mtc_40.seq

*The data is located in the provided GitHub repository (https://github.com/momentum-laboratory/deep-molecular-mrf) under a folder named 'supplementary/published_pulse_sequences'. All .seq format files were also deposited in the pulseq CEST open library[80].

**Previously published pulse sequences in .seq format for[32] and[29] are available on the pulseq CEST open library https://github.com/kherz/pulseq-cest-library/tree/master/seq-library[80].

Table 2. Troubleshooting

Step	Problem	Possible reason	Solution

SWIG installation	SWIG-related error	No SWIG installed	Linux machine:
			
			`apt-get update`
			`apt-get install -y gcc g++ libstdc++6 swig`
			
			Windows/Mac:
			Install using the link https://www.swig.org/

CEST-MRF package installation	Errors related to BMCSimulator or cest_mrf package installation	No SWIG installed or simulation compiler error	If the cest_mrf installation fails, you can delete the re-install the Python=3.9 environment.
			
			If the BMCSimulator fails, verify that was SWIG installed and follow the instructions for manual compilation:
			
			`cd open-py-cest-mrf/cest_mrf/sim_lib/`
			`python setup.py build_ext --inplace`
			`python setup.py install`
			
			If the problem persists, consider using the provided docker image: (https://hub.docker.com/repository/docker/vnikale/pycest-starter/general) following docker installation: (https://docs.docker.com/engine/install/).

Torch installation	First run in Jupyter requests installing the ipykernel	No ipykernel installed	Install the required package: `pip install ipykernel`

	CUDA related problems	CUDA version used does not match Torch version.
		
		No CUDA toolkit installed.
			First check the version of installed CUDA. In Linux you can run:
			
			`nvidia-smi`
			
			In Windows you can check Nvidia driver. In Mac: check the Activity Monitor.
			
			Next, install a Torch version that matches your CUDA version. (https://pytorch.org/get-started/locally/). Alternatively, you can install CUDA-toolkit in conda, corresponding for the latest version of Torch: (https://anaconda.org/nvidia/cuda-toolkit).
			
			Possible solution for conda env:
			
			`conda install nvidia/label/cuda-11.8.0::cuda-toolkit`
			`pip3 install torch torchvision --index-url https://download.pytorch.org/whl/cu118`

Sequential and deep reconstruction –	OpenCV related errors	No OpenCV installed (required for visualization)	Reinstall opencv. If you are using conda: `conda install -c conda-forge opencv`

mouse imaging case			Otherwise (for Linux):
			
			`apt-get update && apt-get install ffmpeg libsm6 libxext6  -y`
			`pip install opencv-python`

Table 3. Timing.

Stage	Step	Timing
In-vitro phantom preparation (**optional**)	3h	
Pulse sequence setup	CEST MRI simulator installation	5-10 min
	Loading and installing the pulse sequences into the MRI scanner	5 min
	CEST MRF pulse sequence file creation	30 sec
Animal / human subject setup	Animal / human subject setup	10 min per animal / 5 min per subject
Raw data acquisition (per subject)	Semisolid MT/ CEST MRF acquisition	35 s – 5 min
	$T_1/T_2/B_0/B_1$ map acquisition(**optional**)	8-15 min
	CEST Z-spectrum acquisition (**optional**)	3 - 10min
Image analysis, processing, and AI-based reconstruction	CEST MRF dictionary simulation	10 sec - ~22 hrs*
	Dot-product matching	10 sec - 2.3 hrs*
	Supervised reconstruction network training	5 min - 6h
	Supervised reconstruction network inference (including variable preparation and loading)	10 sec
	Sequential reconstruction network training and inference	10-15 min
	Multi-pool inference of human parameter maps (including variable preparation and loading)	10 sec
	Self-supervised learning (**optional alternative**)	10 sec - ~22 hrs**
Total protocol time	Depends on the imaging scenario and optional steps	48 min - 57 hrs

Timing was performed on a desktop computer equipped with an Intel I9-12900F CPU, 32GB RAM, and an Nvidia RTX 3060 12 GB GPU.

*The timing is given for L-arginine phantom data (CW pulses with ~665K dictionary entries). Changing the imaging scenario of interest may result in drastic timing changes, depending on the dictionary size, saturation pulse properties (CW/PW) and the number of simulated pools. For example, generating a dictionary of 273,790 entries, acquired using a PW clinical sequence (see the dp_clinical case folder in the provided GitHub repository) took 190s. A previous work reported the generation of ~70M entries for a multi-pool pulsed sequence in ~22 hours and the corresponding dot-product matching (for a single slice image) taking ~2.3 hours[29].

** The inference takes about 10 s. A previous work reported that training using 5 subject data and 153 slices took ~22 hours[40].